



Smart Helmet based Accident Detection and Notification System for Two-Wheeler Motor Cycles

Team Members Details

Mr. MUKIL S (22BEC0480)

Mr. NARAYANAN R (22BEC0492)

Ms. MOHAMED SAFIQ S (22BEC0497)

Project Guide:

Dr. AARTHI G
Associate Professor

Outline:

- Novelty of the work
- Functional Block Diagrams
- Implementation Detail
- Development Status
- Hardware/Software Details
- Photos/Screenshots
- Results
- Timeline plans
- References

Novelty of the work

1) HOLISTIC SAFETY SYSTEM

Unlike existing works that address only one feature such as accident detection or helmet enforcement, this project develops a single, integrated solution that combines helmet use verification, alcohol detection, navigation assistance, and accident response.

2) DUAL PRE-RIDE SAFETY ENFORCEMENT

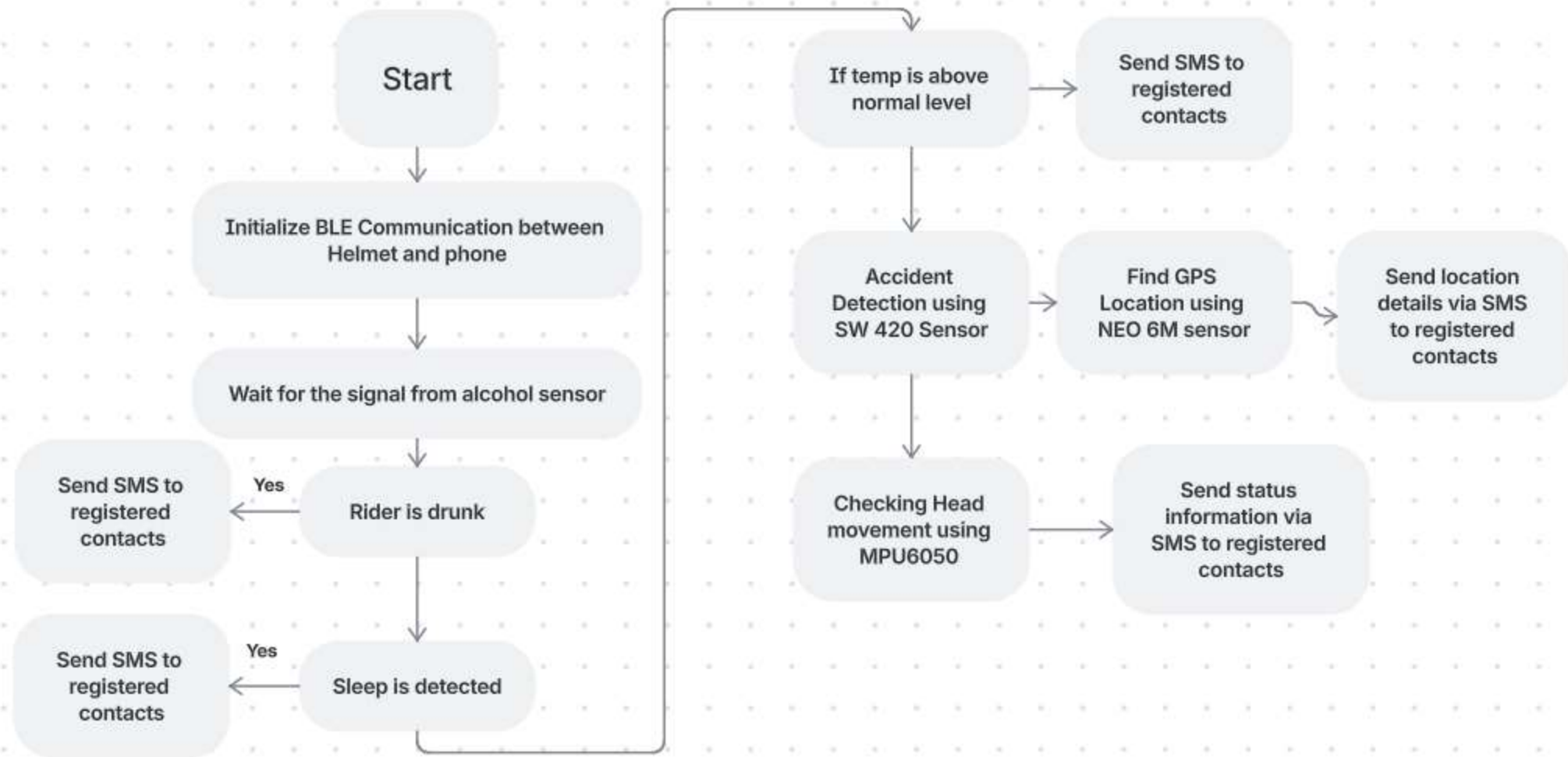
Most studies focus on either helmet enforcement or alcohol detection.

This project is unique in making both checks mandatory before ignition, ensuring maximum rider safety right from the start.

3) ENHANCED EMERGENCY RESPONSE & NAVIGATION

The system not only detects accidents in real time and sends precise GPS-based alerts to emergency contacts, but also provides distraction-free navigation cues via helmet LEDs, improving both post-accident response and on-road safety.

Functional Block Diagram:



Implementation details

Components used:

- ESP 32
- MQ3(alcohol sensor)
- SW 420(accident detection)
- Sleep detection using IR sensor
- Neo-6m(GPS coordinates)
- DHT 11(Temperature Sensor)
- MPU6050(Head movement detection Sensor)

Software Platforms:

- Arduino IDE
- Android studio

Development Status:

- A prototype **Smart Helmet** has been developed using the **ESP32 microcontroller** to enhance rider safety through real-time monitoring and alert mechanisms. The system integrates multiple sensors to detect unsafe riding conditions and respond effectively.
- An **MQ-3 alcohol sensor** is used to measure alcohol concentration in the rider's breath. Dynamic threshold calibration ensures reliable detection under varying environmental conditions. If alcohol consumption exceeds the safe limit, the system triggers an alert and prevents unsafe riding.
- The **eye blink sensor** continuously monitors the rider's eye movement to detect drowsiness or sleep while riding. When prolonged eye closure is detected, the system identifies fatigue and issues a warning to prevent accidents caused by driver inattention.
- An **SW-420 vibration sensor** is employed to detect sudden impacts or abnormal vibrations, indicating a possible accident. Upon detection, the system automatically generates an accident alert.
- To enhance safety further, a **GPS module** is integrated to track the real-time location of the rider. In case of an accident or alcohol detection, the system sends the exact geographical coordinates to a predefined emergency contact for quick assistance.
- Additionally, a **DHT11 temperature sensor** is used to monitor the surrounding temperature, helping in environmental analysis and future safety enhancements.
- The system also includes an **MPU6050 accelerometer and gyroscope** to monitor head movement and orientation. This helps in detecting abnormal head tilts or sudden movements, which can indicate a fall or loss of balance.
- In future, the system will use a GSM module (SIM800L) with a 2G SIM card to send SMS alerts containing GPS coordinates when an accident is detected.

Hardware/Software Details:

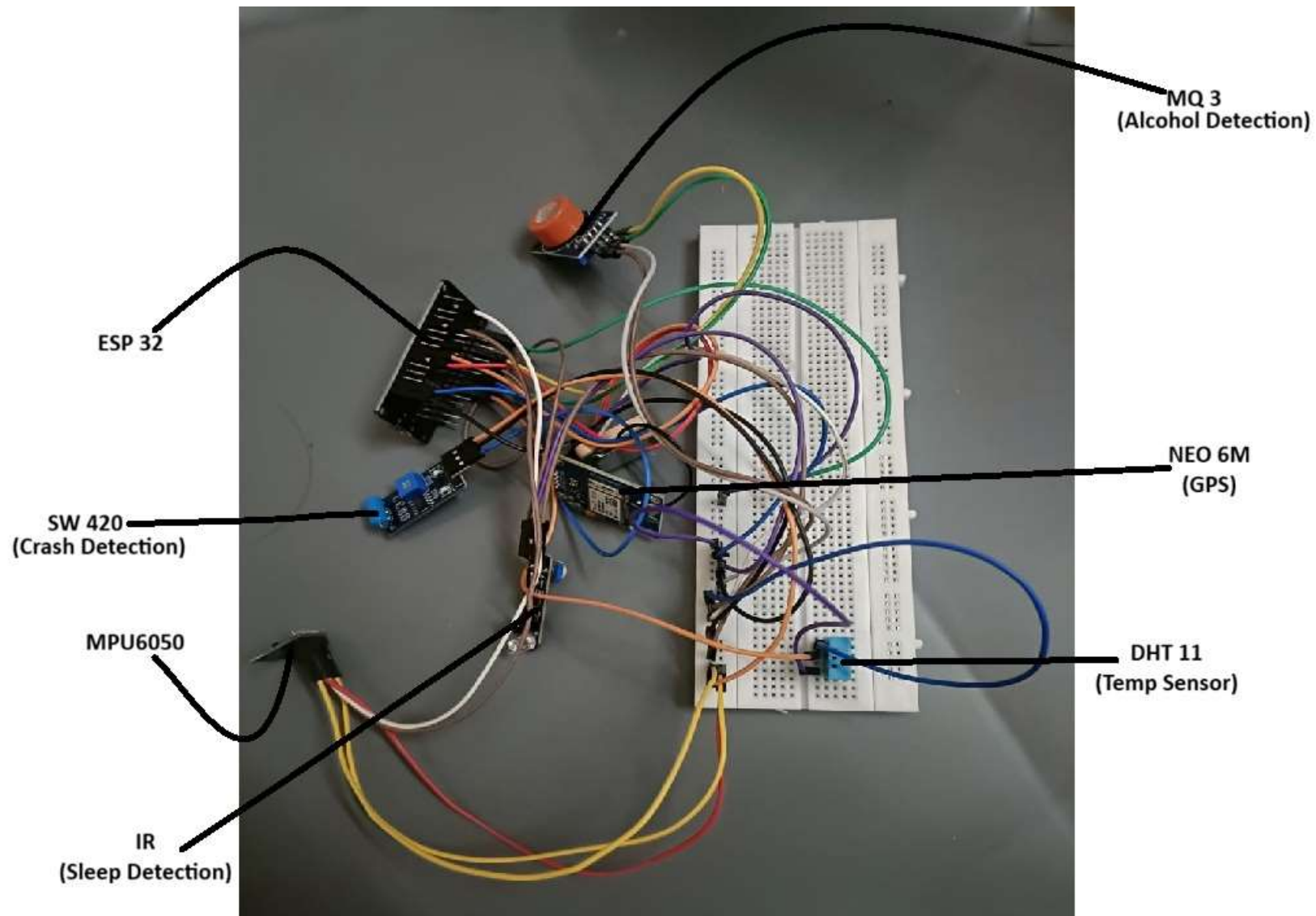
Hardware

- ESP 32
- MQ3(alcohol sensor)
- SW 420(accident detection)
- Sleep detection using IR sensor
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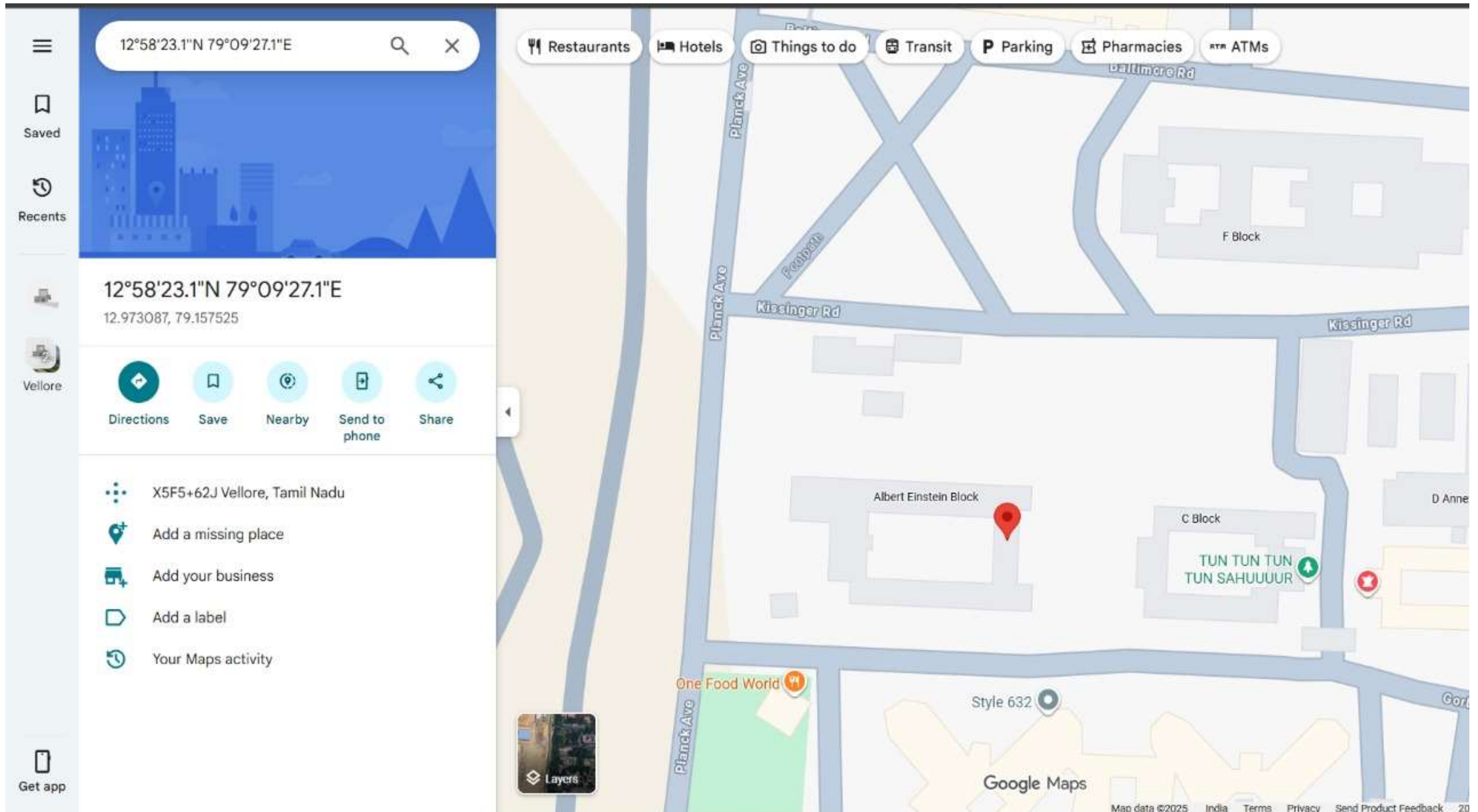
Software

- Arduino IDE
- Android Studio

Photographs/Screenshots



Photographs/Screenshots



Results

Output Serial Monitor X

Message (Enter to send message to 'ESP32 Dev Module' on 'COM3')

```
17:24:16.240 -> 🚀 SMART HELMET SYSTEM STARTED
17:24:16.273 -> 📡 Calibrating MQ-3...
17:24:19.239 -> 📡 Alcohol Threshold: 1921
17:24:19.272 -> 👁 Calibrating Eye Sensor (Eyes Open)...
17:24:23.227 -> 👁 Eye Threshold: -270
17:24:23.256 -> ✅ RIDER IS SOBER
17:24:23.289 -> 🌡️ Helmet Temp: 26.80
17:24:39.838 -> 💥 ACCIDENT DETECTED
17:24:39.870 -> 🗣 Monitoring head movement after impact...
17:24:40.038 -> 🟡 PRIORITY: MEDIUM (Impact + Movement)
17:24:40.103 -> 🚒 PRIORITY LEVEL: MEDIUM
17:24:40.135 -> 📍 Fetching GPS location...
```

Timeline: (Mid July to Mid November)

S.No	Task Description	Time schedule of the progress			
		1-4 Weeks	5-8 Weeks	9-12 Weeks	13-16 Weeks
1	Planning & Setup: Selected and tested MQ-3, SW-420, Eye Blink, GPS, and ESP32 modules				
2	Integration: Connected all sensors to ESP32 with alcohol detection and GPS tracking.				
3	Feature Expansion: Added sleep and accident detection with alert and location logging				
4	Testing & Upgrades: Tested prototype, improved accuracy, and planned IoT/SMS and ignition interlock features.				

Note: The Timeline mentioned in this presentation can be customized based on the phases of your project.

References:

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Thank You